

Advanced Lesson 6 - Exercises:

Replacement Rules and Evaluation

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MATH 7013 - Advanced Mathematical Modeling

1: $x = 2$ followed by an integration in x

```
Clear[x];  
x = 2;  
 $\int x^2 dx$ 
```

2: $x := 2$ followed by the same integration

```
Clear[x];  
x := 2;  
 $\int x^2 dx$ 
```

3: A global x colliding with a pattern variable named x

```
Clear[x, f1, f2];  
x = 2;  
f1[x_] = x^2;  
f2[x_] := x^2;  
?f1  
?f2  
{f1[z], f2[z]}
```

```
Plot[{f1[z], f2[z]}, {z, 0, 3}, PlotStyle -> {Thickness[0.01`]}]
```

4: A replacement rule and an integration in the wrong variable

```
Clear[x, y];
```

```
x2 /. x -> y
```

$$\int x^2 \, dy$$

5: Assigning the result of a self-replacement

```
Clear[x, y, z];
```

```
x = x /. x -> y
```

$$\int x^2 \, dy$$

6: The same shape, with a symbol the rule does not mention

```
Clear[x, y, z];
```

```
x = z /. x -> y
```

$$\int x^2 \, dy$$

7: Why two rules in one list do not chain

```
Clear[x, y];
```

```
Sin[x] /. {Sin[x] -> Log[x], x -> 2 π y}
```

```
Sin[x] //. {Sin[x] -> Log[x], x -> 2 π y}
```

```
Sin[x] /. Sin[x] -> Log[2 π y]
```

8: A rule set carried inside a function definition

```
Clear[f, x, y, z];
```

```
f[x_] := x /. {y -> Cos[y], z -> Sin[z]}
```

```
f[x]
```

```
f[x + y + z]
```

9: Frame -> False and Frame -> True as replacement rules

```
Clear[x];
Plot[Sin[x], {x, 0, 2  $\pi$ }, Frame -> False]
Plot[Sin[x], {x, 0, 2  $\pi$ }, Frame -> True]
```

10: Rewriting a finished plot by rewriting its options

```
Clear[x, plot1, plot2];
plot1 = Plot[Sin[x], {x, 0, 2  $\pi$ }, Frame -> True]
?plot1
Frame /. Options[plot1]
plot2 = plot1 /.
  (Frame -> {{True, True}, {True, True}}) -> Frame -> {{False, True}, {False, True}};
Frame /. Options[plot2]
Show[plot2]
```

11: DSolve, substituting constants, checking, and plotting

```
Clear[y, t, sol, sol2];
sol = DSolve[y''[t] + y[t] == 0, y[t], t]
sol2 = sol /. {c1 -> 4, c2 -> 7}
rule = First[sol2]
drule =  $\partial_{\{t,2\}}$  rule
Simplify[y''[t] + y[t] /. drule /. rule]
ysol = y[t] /. rule
Plot[ysol, {t, 0, 2  $\pi$ }]
```